



Module 16

Pacing

Pacing is a key part of running and is perhaps one of the most important things to master, as it can be the difference between finishing strong and dropping out.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Energy allocation
- Relationship between glycogen levels and pacing
- Pacing: An issue of the mind or body?
- Risk management
- Pace-assessment tools
- Pacing strategies
- Pacing variables
- Utilizing lactate threshold training zones for pacing

❖ OVERVIEW

The pacing strategy for a short race (5K) versus a longer race (marathon) is quite different. This is often a difficult concept to grasp for new runners. Push too hard, too early, and pay for it later in the event. The best way to appreciate pacing is to consider the cost/benefit ratio. Too high an energy expenditure (cost), and the benefit (result) will decrease overall. Too low of an energy expenditure, and the result will decrease as well. In order to perform optimally, the energy expenditure must be not too high and not too low. In other words, being an efficient runner is a function of balance.

Race pace is determined during training and based on assessments of a client's fitness level. Correlating a client's effort level, such as LT to assessment methods (i.e., heart rate, power, RPE, etc.), will enable the person to track pace throughout an event.

For example, while a client might want to break three hours for a marathon, if the individual can average only a 10-min/mile, this is not a feasible goal. Assuming a client has a time goal, it must be realistic.

When participating in a race, unplanned events can occur, such as bathroom breaks, cramping, and falls, just to name a few. It is imperative that your client not exceed the planned pace after being delayed. Deviating from the planned pace almost always ends in a slower overall time.

Smooth and steady is always the better pacing option than erratic and punchy. The more erratically runners pace themselves, the less efficient they will be. For example, while it might seem faster to sprint up the hills and recover on the way down during a hilly race, this will likely result in a slower time than a competitor who keeps a constant effort. This does not mean that an athlete should not push harder in some sections of a course, but the efforts need to be calculated to ensure the overall pace is sustainable for the duration of the race.

While specific to cycling, a study that examined power output during Olympic distance triathlons found that large power variations equated to a higher overall workload versus cycling at a constant power output (470). Additionally, a study by Rapoport (705) found that running at a constant pace is the most efficient pacing strategy. However, Rapoport noted that in order for running at a constant pace to be the most efficient strategy, the terrain must not vary. Most outside courses

have some sort of terrain variation. In these cases, the most efficient way to run is not at a set pace, but rather at a set level of exertion or effort.

It is important that you know the course profile of a race so that you can plan a detailed pacing strategy for your client.

Figure 16.1 shows various pacing strategies. Of the four strategies, only the blue and orange are recommended.

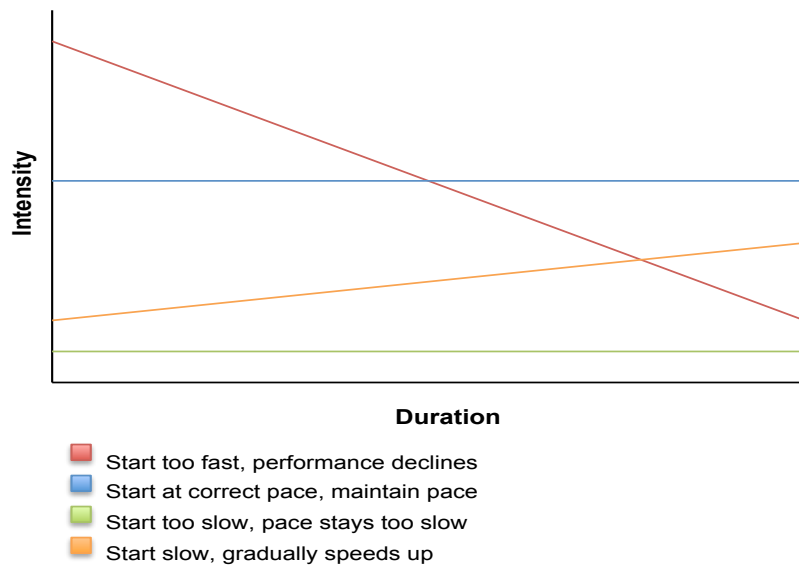


Figure 16.1 Pacing strategies

IMPACT ON GLYCOGEN IN RELATION TO VARIED RUNNING SPEED

To run a predetermined distance in a set time, a runner has to maintain a specific average speed for the entire course. If a runner dips below this average pace, the individual must run faster than the average at some point to make up for going slower than the average pace. You may be thinking, *“Big deal, a little slower here, a little faster there ... so long as I average the same overall pace, what does it matter?”*

From the standpoint of glycogen utilization, even if the finishing time is the same (i.e., same average pace) between a runner who maintains the same pace and one who varies pace, there is a greater metabolic cost because of increased

glycogen consumption with a varied pace. This puts a runner at risk of a slower overall time or, in the worst-case scenario, hitting the wall.

Car Analogy

Two cars drive a distance of 65 miles. One drives at a steady speed of 65 mph while the other is constantly braking and accelerating between 40 to 80 mph. While both cars arrive at the finish in one hour, guess which car was more inefficient and thus used more gasoline? The car driving at varied speeds.

This example illustrates the importance of maintaining a steady pace and/or intensity level over the distance of an event.

❖ SELF-REGULATION: MIND OR BODY?

Intensity and fatigue are invariably linked to endurance sports training and, thus, pacing. The longer and/or more intensely an individual continues to exercise past the initial sensation of muscle tightness, the more fatigued the person will become. If exercising to an extreme level, eventually the level of performance will decrease, and if continued long enough, exercise cessation will likely occur.

The human body regulates itself to ensure survival and to operate efficiently. What is up for debate is the exact means by which the body regulates itself and to what extent.

FATIGUE

Fatigue is a catchall term that is used in many different contexts. For example, fatigue can be used to denote mental or physical stress. Additionally, the exact definition of fatigue is also up for debate. Some view fatigue as a progressive process that begins at the onset of exercise while others view it as the involuntary end point of an exercise bout (542, 554). Causes of physical fatigue likely differ based on the mode of exercise and its duration and intensity.

Physiological changes that occur within a motor unit in regard to muscular fatigue are termed **peripheral fatigue (PF)**. This is considered the traditional model of fatigue. For example, if when running the quadriceps begin to “burn” and fatigue, this is indicative of a buildup of acidity within the muscle, which slows the runner and possibly causes the runner to stop. In other words, the physiological changes that result in fatigue are isolated to specific, local muscle motor units (545). To review, a motor unit comprises a motor neuron and the muscle fibers that the neuron influences (innervates). Below is an example of peripheral fatigue.

CAUSE: Running uphill

RESPONSE: Buildup of intramuscular acidity – causing muscle burn

EFFECT: Runner slows or stops to recover

The opposing model of fatigue is termed **central fatigue (CF)**. Central fatigue is the result of events within the brain and spinal cord (541). Central fatigue and, more specifically, the **central governor model (CGM)** are discussed in the next section.

CENTRAL GOVERNOR MODEL

The central governor model (CGM) is the opposite of the peripheral fatigue model. Peripheral fatigue states that the muscle itself is the cause of the fatigue, whereas the CGM states the brain, not the muscle, is the reason for a decrease in the performance of a muscle (534, 538). This means that the brain is constantly assessing the body to make sure that we don’t hurt ourselves. In regard to the functionality of the body, the baseline is called **homeostasis**. Homeostasis is defined as (535):

Any self-regulating process by which biological systems tend to maintain stability while adjusting to conditions that are optimal for survival.

The brain monitors the body to ensure that no aspect of the body’s functioning deviates too far from homeostasis. In essence, the brain is the body’s control room and thus its safety mechanism. If the brain senses that something is getting to a potentially dangerous level, it acts to decrease the intensity level of that specific area.

In the example used previously regarding a buildup of intramuscular acidity causing a muscle's function to decrease – under the CGM, this is the result of the brain reducing muscle fiber recruitment of the quadriceps (536).

The CGM was originally proposed by physiologist A.V. Hill in 1924 (537, 539). Hill proposed that the heart was protected by some sort of a governor, likely either from the heart itself or the nervous system (539). More recently, the existence of a central governor was theorized by physiologist Tim Noakes (540, 544). The majority of the information discussed today regarding a central governor is based on the research by Noakes et al.

Specifically, Noakes et al. summarize the CGM as follows:

“Exercise performance is regulated by the central nervous system specifically to ensure that catastrophic physiological failure does not occur during normal exercise in humans.” (542)

“The subconscious brain sets the exercise intensity by determining the number of motor units that are activated and hence the mass of skeletal muscle that is recruited throughout the exercise bout.” (543)

CGM in regard to pacing:

“The brain determines the number of active motor units based on a pacing strategy that will allow completion of a task in the most efficient way while maintaining internal homeostasis and a metabolic and physiological reserve capacity.” (543)

The existence of a central governor in respect to the CGM likely has ramifications regarding the definition of many physiological events and terms. By definition, VO_2 max represents an individual's maximal capacity to transport and use oxygen during exercise. However, under the CGM, an individual's VO_2 max does not represent the maximal capacity to transport and use oxygen, but rather the central governor determines this (602). While some may view this difference in definition as a nuance, physiologically speaking, the distinction is substantial because of its implications.

A study by Karlsson and Saltin noted that performance decreased prior to complete glycogen depletion (716). There is very limited evidence to support

physiologically why this occurs. As such, a decrease in performance prior to total glycogen depletion may be associated with CGM.

Key Aspects of the Central Governor Model

1. The CGM can be thought of as a modified central fatigue model, as the CGM does not identify the spinal cord or motor unit as having a regulatory role (538).
2. The CGM does not identify that changes within a motor unit (i.e., decrease in force production) can occur as the primary means of fatigue. In other words, the CGM views peripheral fatigue as a byproduct of central fatigue (538).
3. The CGM definition of fatigue is not quantitative but rather based on one's sensations (543). This is akin to using rate of perceived exertion (RPE) to assess fatigue.
4. The CGM theorizes that the subconscious brain determines a metabolic cost that is required to perform a physical task (543).
5. Homeostasis is maintained under all conditions of exercise (542).
6. CNS regulates motor-unit recruitment to ensure that ATP production and utilization are matched (538).

Challenges to the Central Governor Model

While the CGM raises many interesting propositions and theories, according to Henriette van Praag, Ph.D., a researcher at the National Institutes of Health, the CGM *“lacks a clear structural/physiological basis within the central nervous system. Thus, empirical evidence for the existence of a ‘governor’ remains to be established.”* (546)

While there is no direct quantitative evidence to prove the existence of a central governor, does it mean that it does not exist? As noted earlier, the CGM is a modified version of the central fatigue model, and thus, it is likely that aspects of the CGM exist to regulate human performance.

Perhaps the most extensive critical review of the CGM is by Weir et al. (538). While the paper identifies many areas of the CGM that Weir and his team feel are not valid in regard to proving the existence of the CGM, Weir et al. conclude that it is likely that both the central and peripheral fatigue models work together to influence and regulate human performance.

Below are some aspects of the CGM that Weir et al. challenged in regard to its validity.

1. While the CGM assesses fatigue based on one's perceived exertion levels, proponents of the CGM assess fatigue by quantitative means (e.g., VO_2 max, blood lactate). Therefore the results are assessing exercise performance, not fatigue (538).
2. The CGM addresses a singular aspect of fatigue whereas in the real world, there are many limiting factors of performance such as motivation and environmental issues (538)
3. According to the CGM, homeostasis is maintained under all conditions of exercise. By definition, homeostasis is not maintained during exercise as varied exercise intensities exist and are likely necessary to stimulate the required physiological adaptations to induce advancements in fitness levels (547).
4. Changes in motor-unit recruitment and firing rate can be influenced by reflex effects of the spinal cord (548, 549). This is at odds with the CGM.
5. During voluntary exercise, fatigue can occur that is not the result of CNS limiting motor-unit recruitment (550, 551, 552, 553).

Discussion

When discussing the peripheral and central governor models in regard to the origin of fatigue, it is essentially the classic argument of which came first, the chicken or the egg. In regard to fatigue, it is likely that both the chicken and the egg occur at the same time. In other words, based on the studies reviewed in this section, it is likely that both the central and peripheral fatigue models coexist to monitor fatigue and, thus, regulate exercise intensity.

Specifically in regard to the CGM, while there are studies that seem to conflict with the existence of the model in its entirety, it is likely that many aspects of the CGM are valid as no fatigue model or theory explains all of the aspects of fatigue (538). More research is required in regard to the existence of a “governor” and, more specifically, what exactly the term “governor” implies and encompasses. However, as Weir et al. propose, the search for a unifying theory of fatigue is “futile” (538).

Based on the likely contribution of both the peripheral and central fatigue models, the following definition of fatigue is suggested:

Any exercise-induced reduction in the ability to exert muscle force or power, regardless of whether or not the task can be sustained (554, 555).

Application

While there is no absolute proof of a central governor, an individual will pace himself or herself differently when racing a mile versus a marathon, and there is less risk involved with running fast for a mile versus participating in a marathon.

Therefore it is important to understand that the distance of a race will have a large impact on a runner’s pace. However, is the central governor’s self-selected pace always correct? Let’s look at a hypothetical example.

Mary is running a half marathon and she feels that a self-selected 9:15-min/mile pace is correct for her to finish the race with the least chance of anything catastrophic (i.e., muscle cramps, glycogen depletion, etc.) occurring. However, based off your training data on Mary, you believe that she could sustain an 8:50-min/mile pace with fatigue levels no higher than would be considered normal for running a half marathon. This example references that depending on the individual, the self-selected pace target (central governor) may be too low, or in some cases, too high.

While the self-selected race pace of a client should be used as a point of reference, other factors such as formal physiological assessments and past training data (i.e., long-run pace) should also be considered when determining pacing guidelines. In other words, the CGM can likely be overridden to some degree.

❖ STICK TO THE SCRIPT

It is important not to get sucked into racing at another person's pace, but to stick to the planned pace. It is easy to get caught up in the excitement of a race and go too fast. Preparation events will allow your client to get exposure to the excitement of race day, while learning how not to translate that excitement into throwing off the pacing game plan.

Utilizing some or all of the assessment tools noted here during training will ensure that a client is racing at the correct pace on race day.

Although it is important to stick to the script, this applies only when your client is healthy, well rested, and performing at expected fitness level. If your client is having an off day and feels terrible, the individual should reduce intensity below what was originally targeted. Your client must listen to his or her body.

During Training

Many runners enjoy training with others for motivation and social interactions. While this is generally a good idea, a common mistake is to train with groups and do workouts that don't follow a runner's specific training program. Your client must be aware of the group workout plan and pass on it if it does not adhere to plan.

Ease Into It

While a warm-up is advised prior to starting a race, it is also advised to start off on the conservative side in regard to pacing. Typically, only elite/professional athletes have the training to start out at a high level of intensity and maintain that level throughout the event (318). The longer the event, the truer this is.

RISK MANAGEMENT

As with most things related to sports, proper pacing comes down to risk management. Go too easy and a client will likely miss the time goal. Go too fast and a client will likely blow up and miss the time goal. However, with proper pacing, a client can set himself or herself up with the best chance to succeed by managing risk appropriately.

Risk vs. Reward

It is common knowledge that the pace for a marathon is slower than that of a 5K. While there are physiological reasons for this, the mentality of a runner also plays a large part. Specifically, distance will dictate how fast an individual will run. As an example, let's assume that Jessica can run a 5K at an eight-minute-mile pace and has never run a marathon. On a whim, Jessica decides to enter a marathon without any specific training (obviously not a smart idea). Jessica paces herself at a 10-minute-mile pace during the marathon. But why? She has no previous experience to tell her how fast to run, so why did she slow down so much in relation to her 5K pace? The answer: While Jessica knows she can run at least 5K at an eight-minute pace, she believes that given the distance of a marathon, she would not be able to hold that pace throughout without blowing up. This is an example of risk management.

Let's assume that Jessica miraculously (in lieu of her lack of training) makes it to 24 miles at a 10-minute pace without blowing up. She decides since there are only 2.2 miles left to go, she will pick up the pace. Jessica lifts her pace to 8:30 min/mile for the remaining distance. Why didn't Jessica pick up the pace prior to this? For the same reason that she initially determined to run a 10-minute pace at the start – there was too much risk in blowing up. With 2.2 miles left to run, she determined that the potential reward of a faster time was worth the risk. In other words, she estimated that the reward (i.e., picking up the pace and finishing with a faster time) outweighed the risk (i.e., blowing up and not finishing or finishing with a slower time).

This represents that the intensity one runs at is highly correlated to the distance left to run. The farther from the finish, the higher the risk – and the closer to the finish, the lower the risk.

In 2010, a group of researchers (de Koning et al.) sought out a way to quantify this phenomenon. In a paper titled *Regulation of pacing strategy during athletic competition*, de Koning et al. created a formula called the **hazard score** (567).

The hazard score correlates the intensity at which an individual is performing to the distance or time remaining in the exercise bout and thus looks to determine the likelihood that an individual will change pace. The formula is as follows (534):

$$\text{Hazard} = \text{Momentary RPE} * \text{Fraction of the distance remaining}$$

While this certification won't go into the specifics of the hazard score paper by de Koning et al., the primary thing to understand is that the RPE of an individual at a set point in time and the distance remaining of an event influence the pace of a runner (567).

PACE ASSESSMENT TOOLS

- Heart rate monitor
- GPS that gives real-time pace
- Time
- Rate of perceived exertion scale (RPE)

In regard to heart rate and as noted earlier in the certification, *cardiac drift* is a phenomenon that elevates one's heart rate without an increase in other performance metrics. It is natural for cardiac drift to occur while running, especially in the latter parts of a training run or race. Cardiac drift can artificially increase one's heart rate by as much as 10–15 bpm. Therefore, it is important to pace not only by heart rate, but also by RPE.

Sample Workout Structures Using Pace Tools

Using some or all pace assessment tools, you can create structured workout sessions for your client. Following are examples of workout structures using the pace assessment tools:

Heart Rate Monitor: Easy run, keep heart rate between 110 and 130 bpm for 30 minutes.

GPS: Five-mile run, run the first mile at nine-minute pace, then run the next three miles at 6:30 pace, and cool down at nine-minute pace for the last mile.

Time: Run 400 meters in two minutes, jog 100m. Repeat the cycle two more times.

Based on the preceding pace tools, it is very easy to allow numbers and quantitative metrics to rule your coaching practice. It is also easy for runners to obsess about their quantitative stats. While pace assessment tools are extremely valuable and a necessity in the training and racing process, it is also important to unplug from this data stream every now and then. Have your client go for a run purely for the fun of it and check out the scenery versus staring at a GPS watch

the whole time! Unplugging is easiest to accomplish and most practical on easy workout days.

Pacing by Feel

Many experienced runners are able to accurately determine their pace by feel. This comes from lots of experience and correlating their pace to pacing tools and RPE. So how does one become accustomed to pacing by feel? Following is a way to test one's "feel" for pace.

Using a stopwatch, run a set distance at a desired pace (ex: run three miles in 24 minutes – eight-min/mile pace), but look at the time only once the distance has been run. If your client has a real-time GPS watch, the person can guess the pace then look at the watch to see how close he or she was.

While this is just one example of how to test one's ability to feel the pace, the tests must be done with some frequency in order to increase one's proficiency at correctly pacing by feel.

If individuals use a pacing tool(s) (i.e., heart rate monitor) to determine effort level, it is important for them to correlate the pacing tool pace with how they feel. For example, if a client's pacing strategy is to maintain an 8:30-min/mile pace during a race but the person cannot maintain this pace because of exhaustion or injury, the runner *must* slow down. Failure to do so will likely result in a slower overall time at best, and a "Did Not Finish (DNF)" at worst. Learning how to pace by feel is an important aspect to develop and master.

PACE STRATEGIES AND VARIABLES

Long vs. Short

Pacing for long- and short-distance events varies. Short-distance events (e.g., 5K) are typically raced at a higher intensity than long-distance events (e.g., marathon), assuming the fitness level of a client is very good and the individual has trained at high intensities. Short events are often considered 10K or less, and long-distance events are typically considered 10K or longer.

Experience/Fitness Variables

Individuals with race-proven fitness levels can typically perform at a higher intensity than those who are new to the sport or not at a high fitness level. We're not talking about a jump from a Zone 2 effort to a Zone 4 effort – we're talking about a relatively small difference. For example, a new runner might run a half marathon in the low to middle part of Zone 2, whereas an experienced participant who has a high fitness level might race it in the middle to high Zone 2 or even low Zone 3.

Many of the pacing guidelines described in this section are identified in ranges based on LT. Less-fit clients should race toward the bottom of the range and more-fit clients should target a pace in the upper range.

In a study of runners at the IAAF Half Marathon World Championships, the fastest runners more or less maintained their pace throughout the race whereas slower runners saw their pace decrease from 5K onward (657).

Pacing by Time

More than any other metric, most runners shoot for a particular overall time. During training and racing, the total event distance is often broken up into sections, each with a set time goal. From a racing perspective, this is done for the purpose of checking one's progress. For example, during a marathon, your client might have set time goals at the 10K, 13.1 miles (halfway), and 30K (18.6 miles) points.

If your client is not at the targeted time at certain checkpoints, it is important not to lift the pace to try to gain back time. This will typically result in an overall slower time and perhaps a DNF.

The best way for your client to reduce overall time without putting in more effort is to focus on running the tangents of the race course.

Pack Pacing

In the study of the IAAF Half Marathon runners previously noted (657), runners who ran in packs had smaller pace reductions than those who did not run in

packs. Therefore running with others in race scenarios is advised in order to run at or near a specific pace.

Age, Gender, and Speed Variables on Pacing

A 2011 study (March et al.) looked at determinants of pacing pertaining to marathons (658). The findings:

- Older runners maintained a more consistent pace than younger runners
- Women maintained a more consistent pace than men
- Faster runners maintained a more consistent pace than slower runners
 - This finding was consistent in other studies as well (659, 660).

These results were consistent with a 2014 study by Trubee et al. that showed women were better at pacing than men, especially in hot weather. The study also found that elite runners were better at pacing than nonelite runners. Not surprisingly, there was no difference in pacing aptitude between elite men and elite women (659).

Effect of Temperature on Pacing

The ideal temperature for running (especially long distances) is between 50 and 53.6 degrees F (729). Within this range, the temperature does not have a negative effect on a runner's pace. Temperatures above or below this range will likely negatively affect pacing. It is likely that temperatures outside this range have a greater negative effect the longer the distance is.

A table by Magill et al. demonstrates the effect temperature can have on pace (723). Like the aforementioned study, this table assumes that 53 degrees F is the ideal temperature for running. At 90 and 10 degrees, the recommended adjusted pace is 27 seconds per mile slower. For example, if a runner's pace at 53 degrees F is 8:00 min/mile, the pace at 90 and 10 degrees is 8:27 min/mile (723). Small temperature variations from 53 degrees F, such as 60 degrees, have little to no change in recommended pace changes. However, the more extreme the temperature change, the more effect it has on the recommended pace.

Be aware that any temperature/pace guideline chart is just that, a guideline. Individuals' tolerance for temperature changes vary substantially, and adaptations such as heat-acclimation training will lessen the negative effect of heat on an individual.

Effect of Heat on the Central Governor (CG)

In relation to the central governor model, research shows that an increase in core temperature stimulates the CG in regard to anticipatory regulation – meaning, as one's core temperature rises to the point where fatigue would be likely, the body self-regulates to slow a runner, thereby reducing the person's core temperature (733).

Physiology

As an individual's core temperature rises, an increase in blood is directed to the skin to aid in cooling. This results in a decrease of blood flow to the muscles, thereby reducing the amount of oxygen to the muscles and thus likely decreasing performance (520). Additionally, there is a decrease in ATP production at elevated core temperatures (732).

Mitigating the Effects of High Temperatures

In addition to heat-acclimatization training, precooling the body prior to running (i.e., use of ice vest) and ingesting cold fluids have shown to decrease core body temperature, which may result in increased performance (534).

Break It Up

Breaking a large project into smaller, more manageable pieces not only helps in regard to motivation, but also helps to maintain focus and discipline to accomplish a task. This strategy is often used when running and, more specifically, during longer events such as half and full marathons.

A sample strategy for a marathon might be to break the race up into three parts:

- **Start–13.1:** Run at an RPE of 6/10
- **Miles 13.2–22:** Run at an RPE of 7/10
- **Mile 22–Finish:** Assuming you feel good, run at an RPE of 8/10

This strategy can also be used in shorter events.

COURSE PROFILE (ELEVATION)

As noted in the *Training Program* module, utilization of an elevation course profile can be very helpful in determining a pacing strategy for your client. Even if the advised strategy is to maintain a consistent pace, it is important for your client to know where any hills are. Following is a sample profile of a 10K course.

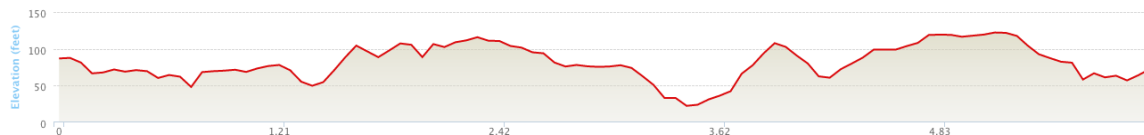


Figure 16.2 Course elevation profile

DETERMINING PERCENT GRADE

Most course profiles equate to little more than a squiggly line and distance/elevation markers. How do you figure out the steepness of the hills? For this you need to think back to middle school and channel your inner 13-year-old – remember “rise over run”?

The equation for this is: $100 \times \text{rise/run}$. As an example, let's say that a road rises 100 feet in one mile (i.e., run). Therefore the equation would be:

$$100 \times 100/5,280 \text{ (feet in a mile)} = \mathbf{1.9 \text{ percent grade}}$$

Sample Race Pace Guidelines

Regarding figure 16.2, a sample pacing strategy might look something like this:

Use miles 0–1.5 as a chance to warm up, maintaining an RPE of no more than 6/10 and target a pace of 8:00 min/mile. Once you hit the hill at 1.5 miles, increase your RPE to 7/10 (note, your pace will drop slightly).

Continue running at this 7/10 pace until you hit mile 3.5, when the big hill begins. Your RPE will likely increase, but try not to go above a 7.5/10 RPE on this hill. Maintain this RPE until mile five. At this point, if you feel good, increase your RPE to 8/10 and target a 7:45-min/mile pace if possible throughout the end.

As this example is based on a six-mile course, the pace and RPE increases are likely more aggressive than would be considered appropriate for a longer event, such as a half or full marathon. Longer events would likely have a more consistent pace (RPE-based) throughout the race.

❖ PACING GUIDELINES

The Start

Most runners have some level of anxiety before a race. Combine this with adrenaline and it's easy to see how your client can get lulled into running a pace faster than initially planned.

The run start is a chaotic and stressful environment. Most individuals will have elevated heart rates because of being nervous and stressed – and this is before the gun goes off! This is called “anticipatory anxiety.” In other words, a client is anxious about the start of a race and experiences many physiological changes typically associated with physical exercise (e.g., increased respiration and heart rate). Once the race starts, the stress and anxiety typically decrease.

Your client should focus on pacing by feel and settling into the planned pace. Clients who find themselves running too fast must slow down immediately and get back on track.

A 2010 study by Hausswirth et al. analyzed how the pace of the first kilometer of a run affected overall run performance (367). The results found that running 5 percent slower than one's 10K race pace resulted in the best overall run time. Conversely, running at 5 percent faster or 10 percent slower than one's 10K race pace resulted in an overall slower time. In other words, too slow or too fast a pace in the first kilometer/mile is likely to negatively affect the overall run time (specifically for distances of 10K and below).

Body Awareness

The ability to pace properly is dependent on body awareness, specifically concerning one's form, breathing rate, and overall fatigue/exertion level.

Form

Being aware of one's form is very important while running. If a client notices form is breaking down, it is advised to slow down and work to regain proper form. Common causes of form breakdown are exhaustion and overexertion. Runners often find that doing a body scan is helpful in checking their form. For example, the client might assess the following areas:

Head: Is my neck vertical or substantially flexed or extended?

Back: Is my back hunched (flexed) or extended?

Foot strike: Is my foot strike at the correct size and rate? Am I shuffling?

Hips: Are my hips static or rotating?

Shoulders: Am I elevating my shoulders, or are they depressed and retracted?

Breathing/Stride Rate

Breathing rate can be correlated to one's stride rate. By taking a breath after a set number of strides, your clients will be able to regulate not only the stride rate, but the breathing rate as well. The ability to be aware of the speed and frequency of the stride/breathing rate will allow your clients to stay on pace.

Once clients have established a pace that they feel is appropriate, they need to get a sense of their stride and breathing rate and how it relates to RPE.

Overall Fatigue Level

If at any time clients become overly fatigued, they must slow down. While some degree of fatigue is normal and expected with running, if your client experiences fatigue to the extent that it could impact their safety or their ability to complete a workout or race, they must slow down or stop.

Default to Body Awareness

Regardless of what a heart rate monitor or another quantitative measurement tool says, if clients are feeling overly fatigued they should slow down. However, this

does not work in reverse, meaning, if clients feel great, they should not exceed the established pacing strategy.

What Is a Negative Split?

A negative split is represented by a distance being broken up into two or more sections, with each successive section being run at a faster pace than the segment before it.

Common Pacing Mistakes

Starting Off Too Fast

This is the most common pacing mistake, as even seasoned runners fall victim to it. It is very easy to start off too fast. This is because of many factors:

- Overly excited
- When starting off, a runner doesn't feel fatigued at a faster pace
- Desire to keep up with surrounding runners

While this can occur at any distance event, it is most common in short races (10K or less).

Running Someone Else's Pace

It is common to run at another person's pace. This is especially true if an individual is running near another runner for long periods and that runner either speeds up or slows down. In regard to slowing down, often this change of speed is not a conscious effort. Lastly, some runners get competitive with other runners during a race and change their speed based on beating those around them.

Running at Varying Intensities

The most common issue in this regard involves hills and descents. Many runners try to maintain the same speed (pace) up hills as they do on the flats. Conversely, many runners relax while going downhill, causing their form to break down. Running at varying intensities is less efficient than running at a constant intensity and therefore typically results in a slower time and potentially a "Did Not Finish (DNF)" because of excessive energy expenditure.

DISTANCE-SPECIFIC PACING GUIDELINES

The following guidelines are rough recommendations, as the exact pacing prescriptions are individually based. Typically, the more in shape and/or advanced a runner is, the longer the person can run at MLSS. Additionally, an in-shape runner can typically run at paces higher than MLSS for longer periods than someone who is not as aerobically conditioned.

Short Distance

Assuming your client is in good physical condition, race distances of 5K and less can be run at a relatively high level of intensity compared with that of long-distance events. Depending on one's fitness level, the advised intensity range is between 95 and 110 percent of one's MLSS.

Events that are around 10K in length are done at a more conservative pace. This typically equates to racing at 90–100 percent of one's MLSS.

Long Distance

For half marathons, the pace should be between 75 and 85 percent of one's MLSS. From an RPE perspective, the intensity should feel like a long-tempo run (368).

Pacing for marathons should be between 70 and 80 percent of one's MLSS. From a heart rate perspective, the pace should be five to 10 bpm lower than a long-tempo run. The RPE should feel like a long endurance-based run.

Comparative Pacing Guideline Charts

These charts estimate one's suggested pace for a set distance by knowing the pace/time for another distance (i.e., I can run a 5:50-min/mile pace for 5K, and therefore the chart says I can run at 6:00 min/mile for a 10K).

While these charts are great to ballpark estimate paces, they should be viewed as just that, an estimate. Given the wide range of individual variables, it is suggested to view the results of these charts with a grain of salt.

Therefore the following pacing guidelines should be used only as a resource.

Benchmarking Paces

A good benchmark to base paces off is the 5K (3.1 miles). Having a client run a 5K as fast as possible will give you a starting point to determine paces for lactate-threshold runs as well as harder-intensity runs such as VO_2 max efforts.

PACE CONVERSIONS

Once you have the time for a 5K, multiple the minutes by 60 and then add the seconds.

Ex: **5:23 pace**

1. $5 \text{ (minutes)} * 60 = 300 \text{ (seconds)}$
2. $300 + 23 \text{ (seconds)} = \mathbf{323 \text{ seconds}}$

Lactate-threshold (MLSS) workouts are an important part of a training program and represent approximately a 7 percent slower pace than that of a 5K (783). Conversely, VO_2 max intensity-pace workouts are approximately 7 percent faster than 5K pace (783). Following are the conversions based on the aforementioned 5K pace (323 seconds).

Maximum Lactate Steady State (approx. 10-mile-race pace)

1. $323 \text{ (seconds)} * .07 = 24.5 \text{ seconds}$
2. $323 + 24.5 = 348 \text{ seconds}$

3. $348/60 = 5.8$
4. The "5" represents five minutes, and now you need to figure out how to convert .8 to minutes.
 - a. $.8/10 = .08$
 - b. $.08 * 600 = \mathbf{48 \text{ seconds}}$
5. Therefore the pace for lactate threshold runs is **5:48**.

VO₂ Max (approx. 2-mile-race pace)

6. $323 * .07 = 24.5$
7. $323 - 24.5 = 299 \text{ seconds}$
8. $299/60 = 5.0$

For the preceding example (VO₂ max), the pace is 5:00, as there are no seconds. However, let's look below at an example that requires seconds to be determined.

5K pace: **7:17**

1. $7 * 60 = 420 + 17 = \mathbf{437 \text{ seconds}}$
2. $437 * .07 = 30.6$
3. $437 - 30.6 = 406$
4. $406/60 = 6.76$
5. The "6" represents six minutes, and now you need to figure out how to convert .76 to minutes.
 - a. $.76/10 = .076$
 - b. $.076 * 600 = \mathbf{46 \text{ seconds}}$
6. Therefore the pace for VO₂ max runs is **6:46** (based on a 5K pace of 7:17 min/mile).

Now that you know how to perform pace conversions, how does this relate to determining paces for workouts and race distances? To some degree, target paces should be determined based on past workouts and races versus a formula. However, below are some ballpark estimates based off a 5K time.

The following diagram denotes multiple event distances and workout types based off a 5K pace of seven-minute miles.

- Easy/recovery runs are typically at or slower than a long-run pace.
- Interval paces are based on the distance but are typically at or faster than VO₂ max pace.

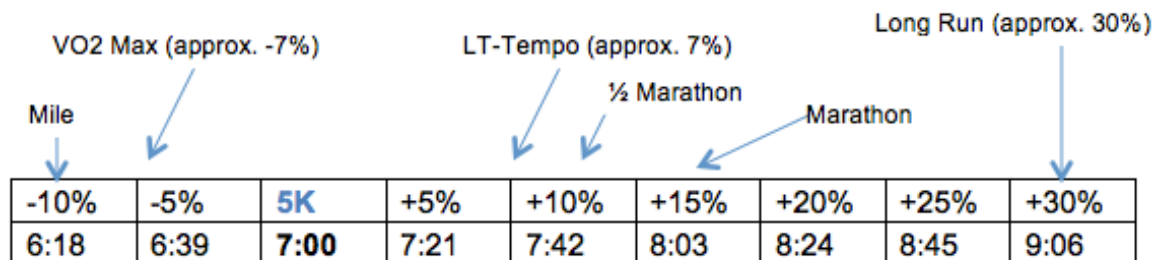


Figure 16.3 Pace benchmarks

❖ SUMMARY

- The general pacing theme is to start conservative and finish strong.
- Race pace is determined during training.
- Intensity benchmarks such as heart rate, GPS pace, and rate of perceived exertion (RPE) are ways to assess one's pace.
- It is important to stick to the predetermined pace plan and not deviate from it and, more specifically, not go over the prescribed pace.
 - The only time it is OK to go above the prescribed pace is toward the very end of a run, and only if a client feels able to do so without "hitting the wall."
- It is common for runners to be nervous before a race. This is especially true for new runners and/or before important events.
- Peripheral fatigue relates to fatigue starting within the skeletal muscle whereas central fatigue corresponds to fatigue controlled by the mind. These issues relate to the *central governor model*.
- Even though short-distance events like 5Ks are short compared with marathons, they are still endurance events and therefore proper pacing is important for a successful race.
- Pacing is typically based either off pacing metrics (GPS), heart rate, RPE, or a combination of them.
- Generally speaking, the shorter the event, the faster the pace.
- Overreaching too early in a distance-running race will reduce the energy and performance later in the race.
- Pacing must be practiced in training to be effective on race day.
- Default to body awareness: Regardless of what a heart rate monitor says, if your client is feeling fatigued the individual should slow down.